

Year 7 • Chapter 1 • Integers: answers (teacher copy)

1.01 Integers and opposites (booklet p. 3)

WE DO examples

- Ex 1: -6 ; $+4$
- Ex 2: 9 , -15 , 0
- Ex 3: -3 , $+3$; back where it started
- Ex 4: -8 (8 units from 0)
- Ex 5: -25
- Ex 6: 7 and -7

Set A

- -5
- 40
- -12
- 8
- -3
- 200

Set B

- 2
- 23
- -56
- 10
- -8
- 0

Set C

- 3 steps back; $+3$ and -3
- $+14$ m (14 m above sea level)
- -7
- -12 (12 units from 0)
- 5 and -5
- You owe \$60 (overdrawn)

Set D

- No, $-0 = 0$
- -18
- -2 , -1 , 0 , 1 , 2
- Yes: every integer has an opposite (0 is its own)

Extension

- 11 ; -11
- No: if n is negative, $-n$ is positive (e.g. $n = -3$, $-n = 3$)

Exit ticket

- Level 1: 15
- Level 2: -9 ; 9
- Level 3: 31 , because the opposite of 31 is -31

1.02 The number line (booklet p. 7)

WE DO examples

- Ex 1: dots at -5 , -2 , 0 , 3
- Ex 2: -2 , -1 , 0 , 1
- Ex 3: dots at -4 , -1 , 2 , 5
- Ex 4: -5
- Ex 5: scale in 5s; dots at -30 , -15 , 10
- Ex 6: 9

Set A

- -7
- -3
- 0
- 2
- 5
- -1

Set B

- dots at -3 , -1 , 1 , 5 , 6
- dots at -7 , -4 , 0 , 4

Set C

- -1 , 0 , 1 , 2 , 3 , 4
- -5 , -4 , -3 , -2 , -1 , 0
- -7 , -6 , -5 , -4
- -6 , -4 , -2 , 0 , 2 , 4 , 6

Set D

- scale -40 to 40 in 10s with dots
- -3 , -2 , -1 , 0 , 1 , 2
- 17
- -3

Extension

- 54
- -6

Exit ticket

- Level 1: -2 , -1 , 0
- Level 2: -8
- Level 3: -3

1.03 Comparing and ordering integers

(booklet p. 11)

WE DO examples

- Ex 1: False; false
- Ex 2: -6 , -1 , 0 , 3 , 5
- Ex 3: $>$
- Ex 4: 8 , 3 , 0 , -2 , -15
- Ex 5: -5 , $-(2)$, 1 , $-(-3)$
- Ex 6: -11 , -10 , -9

Set A

- F
- F
- T
- T
- T
- F

Set B

- $>$
- $<$
- $>$
- $>$
- $<$
- $<$

Set C

- -8 , -2 , 0 , 4 , 7
- 14 , 3 , 0 , -1 , -5 , -9 , -10
- -20 , -15 , -3 , 8 , 12
- 6 , -1 , -2 , -7 , -11
- -12 , -4 , -1 , 0 , 3
- 40 , 12 , -8 , -35

Set D

- e.g. -6 , -7 , -8
- -8
- -1000 , -100 , -10 , -1 , 0
- Yes, for negatives: -20 is smaller than -2

Extension

1. $-4, -(-2), 3, -(-4), -(-9)$
2. e.g. 0 and 2 (any integer from -3 to 2)

Exit ticket

1. Level 1: False
2. Level 2: 2, 0, $-1, -5$
3. Level 3: $-20, -12, 0, 5, -(-8)$

1.04 Adding integers (booklet p. 15)

WE DO examples

1. Ex 1: -1
2. Ex 2: $-10; 4$
3. Ex 3: -11
4. Ex 4: 1
5. Ex 5: 5°C
6. Ex 6: -13

Set A

1. 3
2. 4
3. -7
4. -3
5. -10
6. 3

Set B

1. -5
2. 0
3. -15
4. 14
5. -5
6. -14

Set C

1. -11
2. 1
3. -8
4. 4
5. -11
6. -10

Set D

1. 5°C
2. level 5
3. $-\$15$
4. 6

Extension

1. -12
2. 7

Exit ticket

1. Level 1: 3
2. Level 2: -7
3. Level 3: 1°C

1.05 Subtracting integers (booklet p. 19)

WE DO examples

1. Ex 1: $-4; -12$
2. Ex 2: $7; 4$
3. Ex 3: -12
4. Ex 4: 6
5. Ex 5: 15°C
6. Ex 6: -8

Set A

1. -4
2. -12
3. -7
4. -7
5. -5
6. -20

Set B

1. 7
2. 4
3. 0
4. 10
5. -9
6. 7

Set C

1. -12
2. -7
3. 6
4. -11
5. -5
6. 0

Set D

1. 16°C
2. 25 m
3. -75 m (75 m deep)
4. -13

Extension

1. -8
2. -10

Exit ticket

1. Level 1: -9
2. Level 2: 13
3. Level 3: 19°C

1.06 Multiplying integers (booklet p. 23)

WE DO examples

1. Ex 1: $-18; -10$
2. Ex 2: $48; -24$
3. Ex 3: -42
4. Ex 4: 40
5. Ex 5: -8
6. Ex 6: Positive: 100 negatives make 50 pairs

Set A

1. -18
2. -10
3. -44
4. -63
5. -24
6. -100

Set B

1. 48
2. 25
3. 21
4. 24
5. 20
6. 36

Set C

1. -24
2. -42
3. 40
4. 9
5. 100
6. -8

Set D

1. -28 m (28 m deep)
2. -7
3. Negative: 2 pairs make positive, 1 negative left
4. -18°C

Extension

1. -32
2. 8 and -3

Exit ticket

1. Level 1: -42
2. Level 2: 72
3. Level 3: -60

1.07 Dividing integers (booklet p. 27)**WE DO examples**

1. Ex 1: 3; -3
2. Ex 2: 2
3. Ex 3: -3
4. Ex 4: 8
5. Ex 5: -\$12
6. Ex 6: 54

Set A

1. 3
2. -3
3. 1
4. -4
5. -2
6. -100

Set B

1. -5
2. 7
3. -9
4. 4
5. 0
6. -9

Set C

1. 2
2. -3
3. 8
4. -4
5. 2
6. 8

Set D

1. -\$21
2. -3°C each hour
3. -35
4. -8

Extension

1. -60
2. -5

Exit ticket

1. Level 1: -4
2. Level 2: 7
3. Level 3: -4

1.08 Order of operations with integers

(booklet p. 31)

WE DO examples

1. Ex 1: -12
2. Ex 2: -4
3. Ex 3: -18
4. Ex 4: 7
5. Ex 5: $1 \div (-3) = -1/3$
6. Ex 6: 0

Set A

1. -14
2. -13
3. 0
4. 13
5. 18
6. -3

Set B

1. -18
2. -19
3. -15
4. -2
5. 15
6. 2

Set C

1. 7
2. 0
3. 12
4. 2
5. -1
6. -2

Set D

1. -4
2. $-1/3$
3. 4
4. 4

Extension

1. $(8 - 12) \div (-4 + 2) = 2$
2. e.g. $-3 \times 4 - 5 = -17$

Exit ticket

1. Level 1: 1
2. Level 2: 15
3. Level 3: 12

1.09 Integers on a calculator (booklet p. 35)**WE DO examples**

1. Ex 1: -9; -6
2. Ex 2: 5
3. Ex 3: -12
4. Ex 4: 87
5. Ex 5: -52
6. Ex 6: 4

Set A

1. -9
2. -6
3. -83
4. -48
5. 12
6. -75

Set B

1. -12
2. -52
3. 87
4. -4
5. -7
6. 169

Set C

1. 5
2. 9
3. 36
4. 35
5. -3
6. 36

Set D

1. ≈ -1000 ; -1029
2. ≈ -20 ; -20.5
3. The calculator squares 6 first, then makes it negative.
Enter $(-6)^2$
4. It should be 8: $\times$ is done before $-$

Extension

1. 5
2. -5

Exit ticket

1. Level 1: -45
2. Level 2: 5
3. Level 3: 5

1.10 Integer problems (booklet p. 39)**WE DO examples**

1. Ex 1: 16°C
2. Ex 2: $-\$45$
3. Ex 3: level -3
4. Ex 4: $\$85$
5. Ex 5: $\$820$
6. Ex 6: -12 m

Set A

1. 17°C
2. -4°C
3. -3°C
4. -2°C

Set B

1. $12\text{ m deep } (-11\text{ m})$
2. 5 floors
3. 550 m
4. 155 m

Set C

1. $-\$25$
2. $\$45$
3. $-\$60$
4. $\$150$

Set D

1. $26 \times 80 - 6 \times 210 = \820
2. $35 - 27 = 8$

Extension

1. 70 minutes
2. e.g. It was 4°C and fell 12°C

Exit ticket

1. Level 1: 4°C
2. Level 2: 18 m
3. Level 3: -7